PHYSICS

SECTION-1

1) Which of the following pairs do not match :-

(1) rotational power–Joule/sec
(2) torque-Newton meter
(3) angular displacement-radian
(4) angular acceleration - radian/sec

2) **Assertion :-** A body cannot have mechanical energy without possessing momentum but it can have momentum without having mechanical energy.
Reason :- Momentum and energy have same dimensions.

(1) Both the assertion and the reason are true and the reason is a correct explanation of the assertion.
(2) Both the assertion and reason are true but the reason is not a correct explanation of the assertion.
(3) Assertion is true but the reason is false.
(4) Both the assertion and reason are false.

3) A wheel is spinning at 27 rad/s but is slowing with an angular acceleration that has a magnitude given by ($3t^2$ rad/s^2). It stops in a time of :-

(1) 1.7 s
(2) 2.6 s
(3) 3.0 s
(4) 4.4 s

4)

Two wires of the same length and material but different radii r_1 and r_2 are suspended from a rigid support both carry the same load at the lower end. The ratio of the stress developed in the first wire to that developed in the second wire is :-

(1) $\frac{r_1}{r_2}$
(2) $\frac{r_1^2}{r_2^2}$
(3) $\left(\frac{r_1}{r_2}\right)^{5/2}$

(4) $\frac{r_2^2}{r_1^2}$

5) A force $\vec{F} = 4\hat{i} - 5\hat{j} + 3\hat{k}$ is acting a point $\vec{r} = \hat{i} + 2\hat{j} + 3\hat{k}$. The torque acting about a point $\vec{r}_2 = 3\hat{i} - 2\hat{j} - 3\hat{k}$ is

(1) Zero
(2) $42\hat{i} - 30\hat{j} + 6\hat{k}$
(3) $42\hat{i} + 30\hat{j} + 6\hat{k}$
(4) $42\hat{i} + 30\hat{j} - 6\hat{k}$

6) **Assertion :-** The total energy of a satellite is negative if potential energy is assume zero at infinity
Reason :- Kinetic energy of a satellite is positive.

(1) If both Assertion & Reason are True & the Reason is a correct explanation of the Assertion.
(2) If both Assertion & Reason are True but Reason is not a correct explanation of the Assertion.
(3) If Assertion is True but the Reason is False.
(4) If both Assertion & Reason are False.

7) A solid cylinder of mass m and radius r is rotating about its longitudinal axis (vertical) with angular speed 'ω'. If a disc of mass 2m and radius 2r is gently placed on it coaxially, then the new angular velocity of the system is

(1) ω
(2) $\frac{\omega}{4}$
(3) $\frac{\omega}{8}$
(4) $\frac{\omega}{9}$

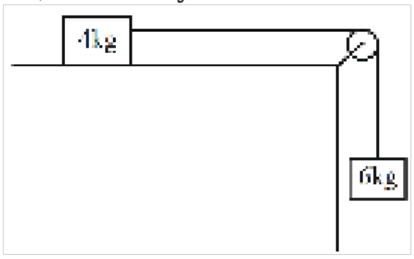
8) Weight of a body of mass m decreases by 1% when it is raised to height h above the Earth's surface. If the body is taken to a depth h in a mine, then its weight will :-

(1) decrease by 0.5%
(2) decrease by 2%
(3) increase by 0.5%
(4) increase by 1%

9) The radius of a wheel of a car is 0.4m. The car is accelerated from rest by an angular acceleration of 1.5 rad/s^2 for 20s. The linear velocity of the wheel is

(1) 10 m/s
(2) 3 m/s
(3) 12 m/s
(4) 2 m/s

10) For the system shown in the figure the acceleration of centre of mass is :-



(1) $\left(2.4\hat{i}-3.6\hat{j}\right)$ m/s^2

(2) $\left(2.4\hat{i}+3.6\hat{j}\right)$ m/s^2

(3) $\left(3.6\hat{i}-2.4\hat{j}\right)$ m/s^2

(4) $\left(3.6\hat{i}+2.4\hat{j}\right)$ m/s^2

11)

Match the column I (rotation of different bodies) with Column II (their moment of inertia) and select the correct answer from the codes given below.

	Column I		Column II
(A)	Thin circular ring of radius R having axis perpendicular to the plane and passing through centre	(1)	$MR^2/2$
(B)	Thin circular ring of radius R having axis passing through its diameter	(2)	$ML^2/12$
(C)	Thin rod of length L about an axis perpendicular to the rod and passing through mid -point	(3)	MR^2
(D)	Circular disc of radius R about an axis perpendicular to the disc and passing through the centre.	(4)	$MR^2/4$

(1) (A) – 3 , (B) -2, (C) -2, (D) -1

(2) (A) –2 , (B) -3, (C) -1, (D) -2

(3) (A) – 3 , (B) -1, (C) -2, (D) -1

(4) (A) – 3 , (B) -1, (C) -2, (D) -4

12) A length of wire increases by 1% on suspending 2 kg. wt. from kit, then linear strain in the wire is :-

(1) $\frac{1}{100}$

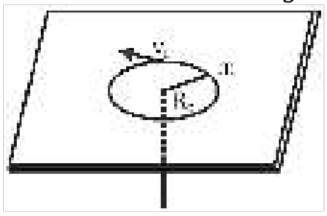
(2) $\frac{1}{10000}$

(3) 1

(4) $\frac{1}{1000}$

13) A mass m moves in a circle on a smooth horizontal plane with velocity v_0 at a radius R_0. The mass

is attached to a string which passes through a smooth hole in the plane as shown.



The tension in the string is increased gradually and finally m moves in a circle of radius $\frac{R_0}{2}$. The final value of the kinetic energy is :-

(1) $\frac{1}{4}mv_0^2$

(2) $2mv_0^2$

(3) $\frac{1}{2}mv_0^2$

(4) mv_0^2

14)

For a constant hydraulic stress on an object, the fractional change in the object's volume $\left(\frac{\Delta V}{V}\right)$ and its compressibility (C) are related as :-

(1) $\frac{\Delta V}{V} \propto C$

(2) $\frac{\Delta V}{V} \propto \frac{1}{C}$

(3) $\frac{\Delta V}{V} \propto C^2$

(4) $\frac{\Delta V}{V} \propto C^{-2}$

15) A thick rope of density ρ and length L is hung from a rigid support. If Y is the young's modulus, the increase in length of rope due to its own weight is –

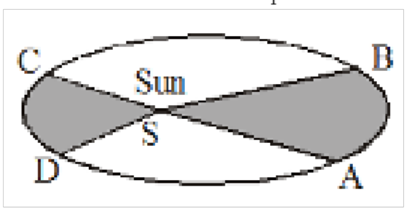
(1) $\frac{1}{4}\frac{\rho g L^2}{Y}$

(2) $\frac{1}{2}\frac{\rho g L^2}{Y}$

(3) $\frac{\rho g L^2}{Y}$

(4) $\frac{\rho g L}{Y}$

16) A planet moves around sun from A to B in time t_1 and from C to D in time t_2. If area ASB is 4 times of that of area CSD then :



(1) $t_1 = t_2/4$

(2) $t_1 = t_2$

(3) $t_2 = t_1/4$

(4) $t_2 = 2t_1$

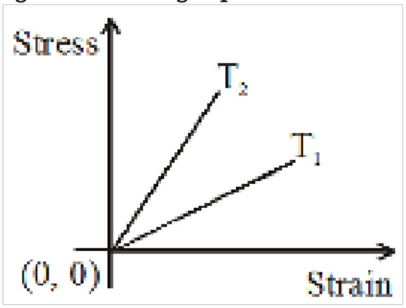
17) The density of a newly discovered planet is twice that of earth. The acceleration due to gravity at the surface of the planet is equal to that at the surface of the earth. If the radius of the earth is R, the radius of the planet would be

(1) 2R

(2) 4R

(3) R/4

(4) (R/2)

18) Figure shows graph between stress and strain for a uniform wire at two different temperatures then 

(1) $T_1 > T_2$

(2) $T_2 > T_1$

(3) $T_1 = T_2$

(4) None of these

19) The acceleration due to gravity g and mean density of earth ρ are related by which of the following relations ? [G = gravitational constant and R = radius of earth] :

(1) $\rho = \dfrac{4\pi g R^2}{3G}$

(2) $\rho = \dfrac{4\pi g R^3}{3G}$

(3) $\rho = \dfrac{3g}{4\pi G R}$

(4) $\rho = \dfrac{3g}{4\pi G R^3}$

20) Internal forces in a system can change :

(1) linear momentum only

(2) kinetic energy only

(3) both kinetic energy and linear momentum

(4) neither the linear momentum nor the kinetic energy of the system

21) A satellite is moving in a circular orbit around earth with a speed 2v, if it's mass is m, then its total energy will be :-

(1) $\frac{3}{4}mv^2$

(2) mv^2

(3) $\frac{1}{2}mv^2$

(4) $-2mv^2$

22) Let $\vec{F}$ be the force acting on a particle having position vector $\vec{r}$ and $\vec{\tau}$ is the torque of this force about the origin then correct option will be :

(a) $\vec{F} \times \vec{\tau} = 0$

(b) $\vec{\tau} . \vec{r} = 0$

(c) $\vec{\tau} \times \vec{r} \neq 0$

(d) $\vec{F} \times \vec{\tau} \neq 0$

(e) $\vec{\tau} \times \vec{r} = 0$

(1) a, b, e

(2) b, c, d

(3) b, c, a

(4) b, a

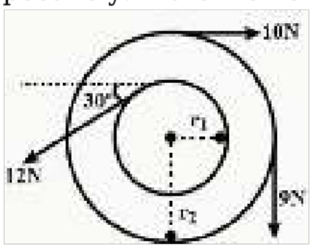
23) A man of mass m starts moving on a plank of mass M with constant velocity v with respect to plank. If the plank lies on a smooth horizontal surface, then velocity of plank with respect to ground is :-

(1) $\frac{Mv}{m+M}$

(2) $\frac{mv}{M}$

(3) $\frac{Mv}{m}$

(4) $\frac{mv}{m+M}$

24) In the following figure r_1 and r_2 are 5 cm and 30cm respectively. If the moment of inertia of the wheel is 5100 kg-m^2 then its angular acceleration will be : 

(1) 10^{-4} rad/s^2

(2) 10^{-3} rad/s^2

(3) 10^{-2} rad/s^2

(4) 10^{-1} rad/s^2

25) Match the Column I with Column II and choose the correct option from the given codes :-

	Column-I		Column-II
(i)	A body which regains its original shape after the removal of external forces.	(p)	Elasticity
(ii)	A body which does not regain its original shape after the removal of external forces.	(q)	Elastic body
(iii)	A body which does not show any deformation on applying external forces.	(r)	Plastic body
(iv)	The property of the body to regain its original configuration when the deforming forces are removed.	(s)	Rigid body

Codes

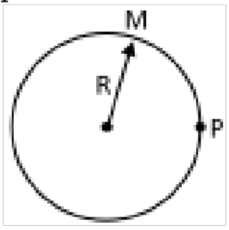
(1) i - q, ii - r, iii - s, iv - p

(2) i - p, ii - q, iii - r, iv - s

(3) i - r, ii - s, iii - p, iv - q

(4) i - s, ii - p, iii - q, iv - r

26) Gravitational potential at point P due to hollow sphere of mass M & radius R is –20 J/Kg then potential at centre O will be :



(1) –10 J/kg

(2) 0

(3) –20 J/kg

(4) ∞

27) A particle of mass 'm' is projected on horizontal ground with an initial velocity of v making an angle θ with horizontal. Find out the angular momentum of particle about the point of projection when it is at its maximum height :

(1) $\dfrac{mv\sin\theta\cos\theta}{2g}$

(2) $\dfrac{mv^3\sin\theta\cos\theta}{2g}$

(3) $\dfrac{2mv^3\sin\theta\cos\theta}{g}$

(4) $\dfrac{mv^3\sin^2\theta\cos\theta}{2g}$

28) If angular momentum of a body is increased by 20% then rotational kinetic energy is increased by (moment of inertia is constant)

(1) 44%

(2) 22%

(3) 40%

(4) 50%

29) Inside a uniform spherical shell :
(a) the gravitational field is zero
(b) the gravitational potential is zero
(c) the gravitational field is same everywhere
(d) the gravitation potential is same everywhere
Choose the most appropriate answer from the options given below :

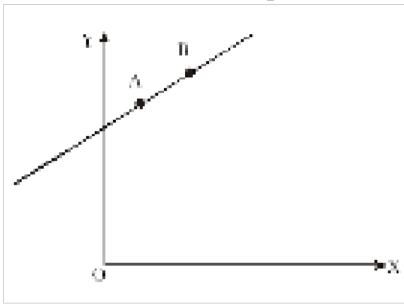
(1) (a), (c) and (d) only

(2) (a) only

(3) (a), (b) and (c) only

(4) (b), (c) and (d) only

30) A particle of mass m moves in the XY plane with a velocity V along the straight line AB. If the angular momentum of the particle with respect to origin O is L_A when it is at A and L_B when it is at B, then :-



(1) $L_A > L_B$

(2) $L_A = L_B$

(3) The relationship between L_A and L_B depends upon the slope of the line AB

(4) $L_A < L_B$

31) Two disc of moment of inertia I and 2I about their respective axes (normal to the disc and passing through the centre), and rotating with angular speeds 3ω and ω respectively in same direction are brought gently into contact face to face with their axes of rotation coincident. The common angular speed of two-disc system is

(1) $\frac{5\omega}{2}$

(2) $\frac{5\omega}{3}$

(3) $\frac{3\omega}{2}$

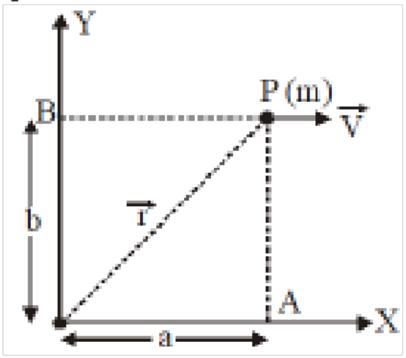
(4) 3ω

32) Every planet revolves around the sun in an elliptical orbit :
A. The force acting on a planet is inversely proportional to square of distance from sun.
B. Force acting on planet is inversely proportional to product of the masses of the planet and the sun
C. The centripetal force acting on the planet is directed away from the sun.
D. The square of time period of revolution of planet around sun is directly proportional to cube of semi-major axis of elliptical orbit.
Choose the correct answer from the options given below :

Options :

(1) A and D only
(2) C and D only
(3) C and B only
(4) A and C only

33) A particle is moving along a straight line parallel to x-axis with constant velocity V. Find angular momentum about the origin in vector form:-



(1) $+mV^2b\hat{k}$
(2) $-mVb\hat{k}$
(3) $-2mVb\hat{k}$
(4) $-mVb\hat{j}$

34) The satellite of mass m revolving in a circular orbit of radius r around the earth has kinetic energy E. Then its angular momentum about the centre of earth is :-

(1) $\sqrt{\frac{E}{mr^2}}$
(2) $\frac{E}{2mr^2}$
(3) $\sqrt{2Emr^2}$
(4) $\sqrt{2Emr}$

35) Copper of fixed volume 'V; is drawn into wire of length 'ℓ'. When this wire is subjected to a constant force 'F', the extension produced in the wire is '$\Delta\ell$'. Which of the following graphs is a straight line ?

(1) $\Delta\ell$ versus $\frac{1}{\ell}$
(2) $\Delta\ell$ versus ℓ^2
(3) $\Delta\ell$ versus $\frac{1}{\ell^2}$
(4) $\Delta\ell$ versus ℓ

SECTION-2

1) A rod of length L is hinged at one end. It is brought to a horizontal position and released. The angular velocity of the rod when it is in vertical position is

(1) $\sqrt{3g/L}$
(2) $\sqrt{2g/L}$
(3) $\sqrt{g/2L}$
(4) $\sqrt{g/L}$

2)

Match the physical quantities given in **Column I** used to describe linear motion with their analogues in rotational motion given in **Column II**.

Column-I	**Column-II**
(A) Mass (m)	(p) Torque (τ)
(B) Momentum (p)	(q) Angular momentum (L)
(C) Force (F)	(r) Moment of inertia (l)
(D) Displacement (Δs)	(s) Angular displacement (Δθ)

(1) (A → r),(B → q),(C → p),(D → s)
(2) (A → p),(B → q),(C → r),(D → s)
(3) (A → p),(B → r),(C → q),(D → s)
(4) (A → s),(B → p),(C → q),(D → r)

3) A constant power is supplied to a rotating disc. Angular velocity (ω) of disc varies with number of rotations (n) made by the disc as :

(1) $\omega \propto (n)^{1/3}$
(2) $\omega \propto (n)^{3/2}$
(3) $\omega \propto (n)^{2/3}$
(4) $\omega \propto (n)^{2}$

4) A man of mass 80 kg stands on a plank of mass 40 kg. The plank is lying on a smooth horizontal floor. Initially both are at rest. The man starts walking on the plank towards north and stops after moving a distance of 6 m on the plank(wrt plank). Then

(1) The centre of mass of plank-man system will move with constant velocity
(2) The plank will slide to the north by a distance 4 m
(3) The plank will slide to the south by a distance 4 m
(4) The plank will slide to the south by a distance 12 m

5) **Assertion (A) :** For a system of particles under central force field, the total angular momentum is conserved.
Reason (R) : The torque acting on such a system is zero.

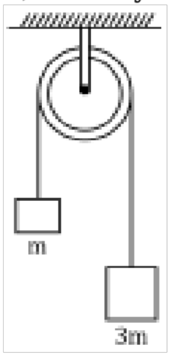
(1) Both A and R are true and R is the correct explanation of A.

(2) Both A and R are true but R is NOT the correct explanation of A.

(3) A is true but R is false.

(4) A is false and R is also false.

6) If the system is released, then magnitude the acceleration of the centre of mass of the system is :



(1) $\frac{g}{4}$

(2) $\frac{g}{2}$

(3) g

(4) 2g

7) Suppose the gravitational force varies inversely as the n^{th} power of the distance. Then, the time period of a planet in circular orbit of radius R around the sun will be proportional to :

(1) R^n

(2) $R^{(n+1)/2}$

(3) $R^{(n-1)/2}$

(4) R^{-n}

8) The angular speed of a motor wheel is increasing from 1200 rpm to 3120 rpm is 16 seconds under the action of constant torque. The angular acceleration is

(1) π rad/s^2

(2) 2 rad/s^2

(3) 2π rad/s^2

(4) 4π rad/s^2

9) A rigid body is rotating such that angular displacement at any instant t is given by $\theta = 2t^2 + 3t + 4$

	Column I		**Column II**
(P)	Angular velocity at t = 2 units	(1)	4 units
(Q)	Angular acceleration at t = 3 units	(2)	0 units
(R)	Average angular velocity between t = 0 and t = 2 units	(3)	7 units
(S)	Average angular acceleration between t = 0 and t = 2 units	(4)	11 units

(1) P→4, Q→1, R→3, S→1

(2) P→3, Q→1, R→4, S→1

(3) P→1, Q→1, R→3, S→4

(4) P→4, Q→3, R→1, S→1

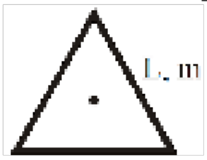
10) Two identical satellites are at the heights R and 7R from the earth's surface. Then which of the following statement is incorrect :- (R = Radius of the earth)

(1) Ratio of total energy of both is 5

(2) Ratio of kinetic energy of both is 4

(3) Ratio of potential energy of both 4

(4) Ratio of total energy of both is 4

11) Three rods each of mass m and length L are joined to form an equilateral triangle as shown in the figure. What is the moment of inertia about an axis passing through the centre of mass of the system and perpendicular to the plane ? 

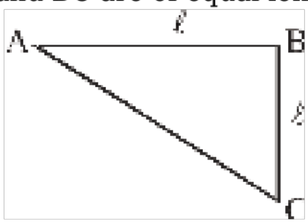
(1) $2\ mL^2$

(2) $\frac{mL^2}{2}$

(3) $\frac{mL^2}{3}$

(4) $\frac{mL^2}{6}$

12) Figure shows a thin metallic triangular sheet ABC. The sides AB and BC are of equal lengths ▯.

The mass of the sheet is M. What is the moment of inertia about AC : 

(1) $\frac{M\ell^2}{18}$

(2) $\frac{M\ell^2}{12}$

(3) $\frac{M\ell^2}{6}$

(4) $\frac{M\ell^2}{4}$

13) Which of the following shows greater increment in length when subjected to same load to wires made of same material :

(1) L = 1 m and r = 1 mm
(2) L = 1 m and r = 2 mm
(3) L = 2 m and r = 1 mm
(4) L = 2 m and r = 2 mm

14) The length of a metal wire is ℓ_1 when the tension in it is T_1 and is ℓ_2 when the tension is T_2. The unstretched length of the wire is :

(1) $\frac{\ell_1 T_1 + \ell_2 T_2}{T_1 + T_2}$
(2) $\frac{\ell_1 + \ell_2}{2}$
(3) $\frac{\ell_1 T_2 - \ell_2 T_1}{T_2 - T_1}$
(4) $\frac{\ell_1 T_2 + \ell_2 T_1}{T_2 + T_1}$

15) The moment of inertia of a solid cylinder about its natural axis is I. If its moment of inertia about an axis perpendicular to natural axis of cylinder and passing through one end of cylinder is $\frac{19I}{6}$ then the ratio of radius of cylinder and its length is:-

(1) 1 : 2
(2) 1 : 3
(3) 1 : 4
(4) 2 : 3

CHEMISTRY

SECTION-1

1)

Trisilyamine $(SiH_3)_3N$ is:

(1) trigonal pyramidal around N and acidic
(2) trigonal pyramidal around N and basic
(3) trigonal pyramidal around N and neutral
(4) trigonal planar around N and weakly basic

2) Possible oxidation states for thallium (TI) are -

(1) 1, 2
(2) 1, 3
(3) 2, 4
(4) 1, 4

3) In the following reaction :-
$B(OH)_3 + H_2O \rightarrow [B(OH)_4]^- + H^+$

(1) $B(OH)_3$ is a Lewis acid
(2) $B(OH)_3$ is a Lewis base
(3) $B(OH)_3$ is amphoteric
(4) None is correct

4) Which of the following carbides will give propyne on reaction with water

(1) Al_4C_3
(2) Al_4C_6
(3) Mg_2C_3
(4) none of these

5) In which of the following compounds B–F bond length is shortest ?

(1) BF_4^-
(2) $BF_3 \leftarrow NH_3$
(3) BF_3
(4) $BF_3 \leftarrow N(CH_3)_3$

6) Boron forms BX_3 type of halides. The correct increasing order of Lewis-acid strength of these halides is

(1) $BF_3 > BCl_3 > BBr_3 > BI_3$
(2) $BI_3 > BBr_3 > BCl_3 > BF_3$
(3) $BF_3 > BI_3 > BCl_3 > BBr_3$
(4) $BF_3 > BCl_3 > BI_3 > BBr_3$

7) Which of the following halides does not hydrolysed?

(1) $PbCl_4$
(2) $SiCl_4$
(3) CCl_4
(4) $SnCl_4$

8) In which of the following reactions, the underlined substance has been reduced?

(1) $\underline{Br_2} + H_2S \rightarrow 2\ HBr + S$
(2) $2\ HgCl_2 + \underline{Sn}Cl_2 \rightarrow Hg_2Cl_2 + SnCl_4$
(3) $Cl_2 + 2\ K\underline{I} \rightarrow 2\ KCl + I_2$
(4) $2\ Cu^{2+} + 4\ \underline{I^-} \rightarrow Cu_2I_2 + I_2$

9) A metal ion M^{3+} loses 3 electrons, its oxidation number will be

(1) +3
(2) +6
(3) 0
(4) –3

10) Oxidation state of each Cl in $CaOCl_2$ is/are

(1) 0 only
(2) +1 only
(3) –1 only
(4) +1, –1

11) Choose the disproportionation reaction among the following redox reactions.

(1) $3\ Mg(s) + N_2(g) \rightarrow Mg_3N_2(s)$
(2) $P_4(s) + 3\ NaOH(aq) + 3\ H_2O(\ell) \rightarrow PH_3(g) + 3\ NaH_2PO_2(aq)$
(3) $Cl_2(g) + 2\ KI(aq) \rightarrow 2\ KCl(aq) + I_2(s)$
(4) $Cr_2O_3(s) + 2\ Al(s) \rightarrow Al_2O_3(s) + 2\ Cr(s)$

12) Which reaction involves oxidation-reduction?

(1) $NaBr + HCl \rightarrow NaCl + HBr$
(2) $HBr + AgNO_3 \rightarrow AgBr + HNO_3$
(3) $2NaOH + H_2SO_4 \rightarrow Na_2SO_4 + 2H_2O$
(4) $H_2 + Br_2 \rightarrow 2HBr$

13) Oxidation number of carbon in C_3O_2, Mg_2C_3 are respectively :

(1) $-\frac{4}{3}, +\frac{4}{3}$
(2) $+\frac{4}{3}, -\frac{4}{3}$
(3) $-\frac{2}{3}, +\frac{2}{3}$
(4) $-\frac{2}{3}, +\frac{4}{3}$

14) Which of the following can act both as an oxidising as well as reducing agent ?

(1) HNO_2
(2) $KMnO_4$
(3) H_2S
(4) H_2SO_4

15) A compound contains atoms A,B and C. The oxidation number of A is +2, of B is +5 and of C is –2. The possible formula of the compound is:

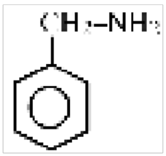
(1) $A_3(BC_4)_2$

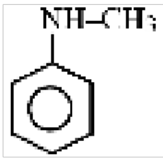
(2) $A_3(B_4C)_2$

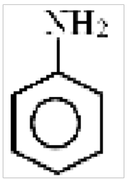
(3) ABC_2

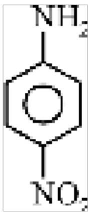
(4) $B_2(AC_3)_2$

16) Which of the following is the strongest base :-

(1) 

(2) 

(3) 

(4) 

17) Which of the following does not react with Na :-

(1) Ph–COOH

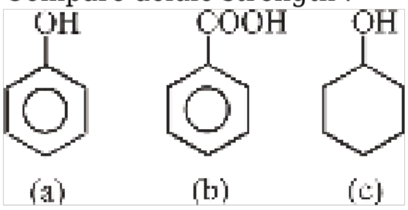
(2) Me–O–Me

(3) Et–OH

(4) Me–C≡CH

18)

Compare acidic strength :-



(1) a > b > c

(2) b > a > c

(3) c > b > a

(4) None of these

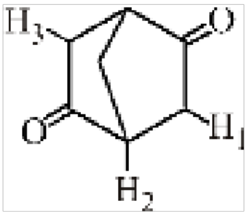
19) Incorrect stability order is :

(1) $Ph_3\overset{\oplus}{C} > Ph_2\overset{\oplus}{C}H > Ph-\overset{\oplus}{C}H_2$

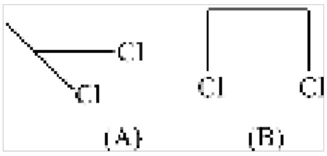
(2) $Ph_3\overset{\ominus}{C} < Ph_2\overset{\ominus}{C}H > Ph-\overset{\ominus}{C}H_2$

(3) $Ph_3\overset{\odot}{C} > Ph_2\overset{\odot}{C}H > Ph-\overset{\odot}{C}H_2$

(4) $Ph_3\overset{\ominus}{C} > Ph_2\overset{\ominus}{C}H > Ph-\overset{\ominus}{C}H_2$

20) Which hydrogen is involve in keto enol tautomerism 

(1) H_1
(2) H_2
(3) H_3
(4) Both H_1 and H_3

21) 

(A) and (B) are :-

(1) Chain Isomer
(2) Position Isomer
(3) Metamer
(4) Functional Isomer

22)

For which carboxylic acid, the pk_a is the lowest:

(1) CH_3-CH_2-COOH
(2) $HC \equiv C-COOH$
(3) $CH_3-CH_2-CH_2-COOH$
(4) $CH_2=CH-COOH$

23) Which of the following statement is incorrect for inductive effect ?

(1) It involves shifting of σ bonded electron
(2) It is distance dependent effect
(3) It arise due to difference in electronegativity of atom attach to organic chain
(4) It is a temporary effect

24) Which of the following has maximum –I effect?

(1) $-NO_2$

(2) $-COOH$

(3) $-F$

(4) $-\overset{\oplus}{N}R_3$

25) Which does not show electromeric effect :-

(1) Acetaldehyde

(2) 2-Butyne

(3) Trans-2-butene

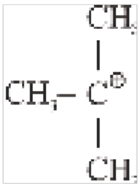
(4) Ethyl methyl ether

26) Most stable carbocation is :-

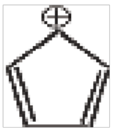
(1) $\overset{\oplus}{C}H_3$

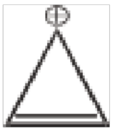
(2) $CH_3-\overset{\oplus}{C}H_2$

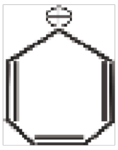
(3) $CH_3-\overset{\oplus}{C}H-CH_3$

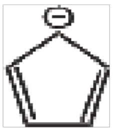
(4) 

27) Which of the following species not an aromatic

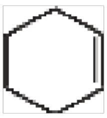
(1) 

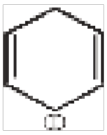
(2) 

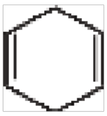
(3) 

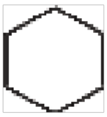
(4) 

28) Which shows resonance :-

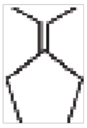
(1) 

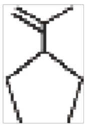
(2) 

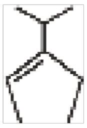
(3) 

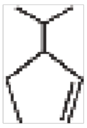
(4) 

29) Which of the following has maximum heat of hydrogenation :-

(1) 

(2) 

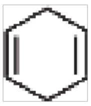
(3) 

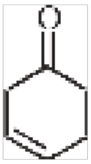
(4) 

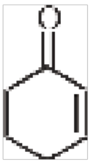
30) Which hydrogen is most acidic :- $\underset{a}{CH_3}\overset{O}{\overset{\|}{C}}\underset{b}{CH_2}\overset{O}{\overset{\|}{C}}\underset{c}{CH_2}\underset{d}{CH_3}$

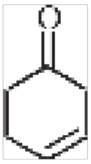
(1) a
(2) b
(3) c
(4) d

31) Conjugation present in :-

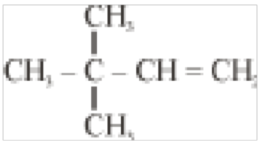
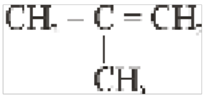
(1) 

(2) 

(3) 

(4) 

32) Which of the following molecule has longest
C=C.

(1) $CH_2 = C = CH_2$
(2) $CH_3 - CH = CH_2$
(3) 
(4) 

33) The compound in which the distance between the two adjacent carbon atoms is largest is :-

(1) Ethane
(2) Ethene
(3) Ethyne
(4) Benzene

34) In Dumas method for estimation of Nitrogen,
0.3 g of an organic compound gave 50 ml of N_2 collected at 300 K temperature and 715 mm pressure. The % of nitrogen in the compound is (aqueous tension at 300 K = 15 mm) :-

(1) 42.5%
(2) 54.3%
(3) 17.4%
(4) 64.7%

35) The blue compound formed in the positive test for nitrogen with Lassaigne solution of an organic compound is :

(1) $Na_4[Fe(CN)_5(NOS)]$
(2) $Na_3[Fe(CN)_6]$
(3) $Fe(CN)_3$
(4) $Fe_4[Fe(CN)_6]_3$

SECTION-2

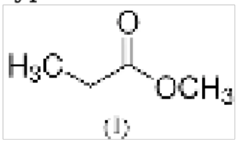
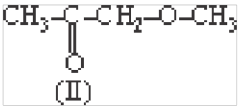
1) The NH_3 evolved from 0.5 gram of an organic compound in Kjeldahl method, neutralized 20 ml of $\frac{M}{2}H_2SO_4$. Find the % of nitrogen :-

(1) 56%
(2) 28%
(3) 14%
(4) 70%

2) The Lassaigne's extract is boiled with conc HNO_3 while testing for the halogens. By doing so it :-

(1) decomposes Na_2S and NaCN, if formed
(2) helps in the precipitation of AgCl
(3) increases the solubility product of AgCl
(4) increases the concentration of NO_3^- ions

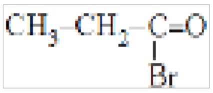
3) Which type of isomerism is observed between

I and II ?  , 

(1) Functional isomerism
(2) Metamerism
(3) Position isomerism
(4) Stereoisomerism

4) Diethyl ether is a metamer of :-

(1) 1–Butanol
(2) 2–Butanol
(3) 2–Methoxy propane
(4) 1–Propanol

5) $BrCH_2–CH_2–CH=O$ and  are

(1) Functional isomers
(2) Position isomers
(3) Chain isomers
(4) Metamers

6) The number of isomers possible with molecular formula $C_2H_4Cl_2$ is.

(1) 2
(2) 3
(3) 4
(4) 6

7)  Relation between above compounds is:

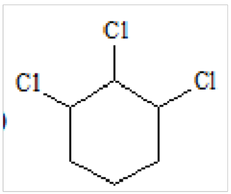
(1) Position isomers
(2) Chain isomers
(3) Identical
(4) Functional isomers

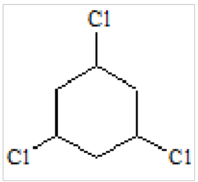
8) C_2H_5CN and C_2H_5NC are

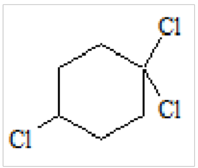
(1) Metamers
(2) Functional isomers
(3) Position isomers
(4) Chain isomers

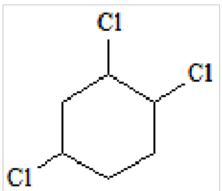
9)

Which of the following compound does not have any geometrical isomer?

(1) 

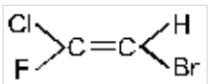
(2) 

(3) 

(4) 

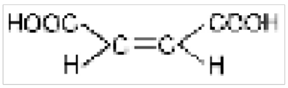
10)

The 'Z'-isomer is :

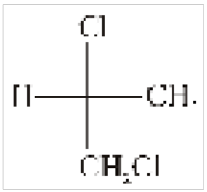
(1) 

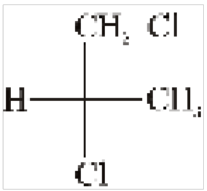
(2) 

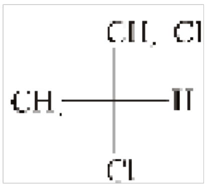
(3) 

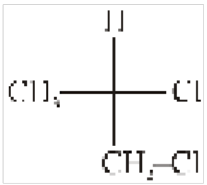
(4) 

11) Which of the following compound has S-configuration :-

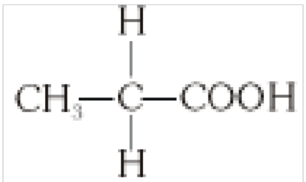
(1) 

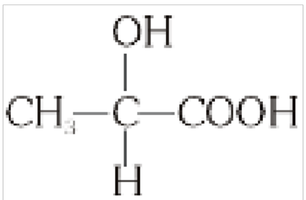
(2) 

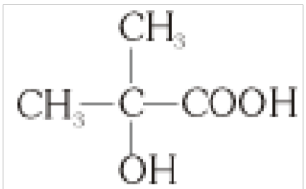
(3) 

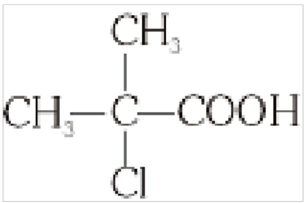
(4) 

12) Which compound is optical active –

(1) 

(2) 

(3) 

(4) 

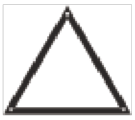
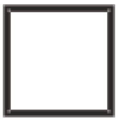
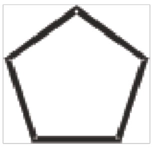
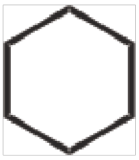
13) Isomers which can be inter converted through rotation around a single bond are:-

(1) Conformers
(2) Diastereomers
(3) Enantiomers
(4) Positional isomer

14) Which conformation of n-butane will have minimum energy.

(1) Gauche
(2) Anti
(3) Fully Eclipsed
(4) Partially Eclipsed

15) Baeyer angle strain per bond is maximum in :-

(1) 
(2) 
(3) 
(4) 

BOTANY

SECTION-1

1) According to IUCN (2004) how many species were made extinct in last 500 years ?

(1) 338
(2) 784
(3) 359
(4) 87

2) Lung of the planet is referred to

(1) Temperate forest
(2) Amazon rain forest

(3) Coral reef
(4) Hot spring

3) In rivet popper hypothesis key species are represented by

(1) Rivets
(2) Rivets on the wings
(3) Entire airplane
(4) Wings of the airplane

4) Robert May, with his conservative and scientifically sound estimate, placed the number of global species diversity at about

(1) 7 billion
(2) 2 billion
(3) 7 million
(4) 2 million

5) The variation in potency and concentration of reserpine present in medicinal plant *Rauwolfia Vomitoria* growing in different Himalayan ranges is due to

(1) Species diversity
(2) Ecological diversity
(3) Genetic diversity
(4) Community diversity

6) Initially how many hotspots were identified globally?

(1) 25
(2) 34
(3) 13
(4) 20

7) Choose the **odd** one for *ex-situ* conservation strategy.

(1) Seed bank
(2) Botanical garden
(3) Wildlife safari park
(4) National park

8) Ecosystem with high species diversity
(a) Is more stable
(b) Is more productive
(c) Is susceptible to disturbances

(1) (a) and (b) only
(2) (b) and (c) only

(3) (a) and (c) only

(4) All (a), (b), (c)

9) Although India has only 2.4 percent of world's land area, its share of the global species diversity is –

(1) 5.1 percent

(2) 6.1 percent

(3) 8.1 percent

(4) 10.1 percent

10) Gametes of threatened species can be preserved in viable and fertile conditions for long periods using

(1) Cryopreservation

(2) *In-situ* conservation

(3) Home gardens

(4) Botanical garden

11) World summit on sustainable development held in

(1) Rio de Janeiro in 2002

(2) Johannesburg 2002

(3) Canada in 2002

(4) Johannesburg in 1992

12) Which among the following is **not** a biodiversity hotspot in India?

(1) Himalaya

(2) Western ghats and Sri Lanka

(3) lndo-Burma

(4) Eastern ghats

13) Most species-rich taxonomic group is of

(1) Mammals

(2) Fungi

(3) Insects

(4) Fishes

14) What is the range of Z (regression coefficient) when analysis is done for the species - area relationships for large area like entire continent?

(1) 0.1 to 0.2

(2) 0.6 to 1.2

(3) 0.1 to 0.6

(4) 0.2 to 0.6

15) India now has __A__ biosphere reserves, __B__ national parks and __C__ wildlife sanctuaries. Select the **correct** option for A, Band C.

(1) A-14, B-90, C-448
(2) A-90, B-14, C-448
(3) A-24, B-89, C-468
(4) A-89, B-24, C-428

16) Alexander Von Humboldt described for the first time

(1) Laws of limiting factor
(2) Species area relationships
(3) Population growth equation
(4) Ecological biodiversity

17) In India, how many genetically different strains of rice and mango varieties are present –

(1) <50,000 and 1,0000 respectively
(2) 1000 and 50000 respectively
(3) >50,000 and 1,000 respectively
(4) >50,000 and 5,000 respectively

18) Leaf base is swollen to form pulvinus in:

(1) Some leguminous plants
(2) Some crucifers
(3) Some monocots
(4) Some cycads

19) The maize grain has an embryo with one large and shield-shaped cotyledon known as

(1) Coleorrhiza
(2) Scutellum
(3) Coleoptile
(4) Epiblast

20) Aleurone layer of maize grain is specially rich in

(1) Proteins
(2) Starch
(3) Lipids
(4) Auxins

21) Placentation, in which ovules develop on the inner wall of the ovary or in peripheral part, is

(1) Free central
(2) Basal
(3) Axile
(4) Parietal

22) Vexillary aestivation is characteristic of the Family

(1) Fabaceae
(2) Asteraceae
(3) Solanaceae
(4) Brassicaceae

23) Drupes are called stony fruits as they have

(1) Stony hard endocarp
(2) Hard mesocarp
(3) Hard epicarp
(4) Hard epicarp and hard mesocarp

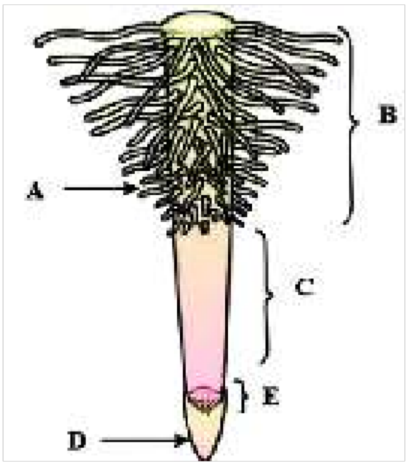
24) Calyx is the outermost accessory whorl of flower. What is the function of calyx?

(1) Helps in pollination
(2) Helps in protection of flower during bud stage
(3) Helps in fertilisation
(4) Helps in seed germination

25) If the margins of sepals or petals overlap one another but **not** in any particular direction, the aestivation is called :-

(1) Imbricate
(2) Valvate
(3) Twisted
(4) Vexillary

26) What indicates A to E in the below figure.



(1) A : Region of maturation, B : Root cap,
C : Region of meristematic activity,
D : Root hair, E : Region of elongation.

(2) A : Root hair, B : Region of maturation,
C : Region of elongation, D : Root cap,
E : Region of meristematic activity.

(3) A : Root cap, B : Region of maturation,
C : Region of elongation, D : Root hair,
E : Region of meristematic acitivity.

(4) A : Region of meristematic activity,
B : Region of elongation, C : Region of maturation,
D : Root hair, E : Root cap.

27) Variation in length of the filament of stamen within flower can be seen in

(1) *Salvia*
(2) Mustard
(3) Chinarose
(4) Both 1 & 2

28) The mode of arrangent of sepals or petals in floral bud with respect to other members of the same whorl is known as

(1) Adhesion
(2) Cohesion
(3) Aestivation
(4) Plancentation

29) In which of the following plant flower can **not** be divided into two similar halves by any vertical plane passing through centre

(1) Mustard
(2) Cassia
(3) Canna
(4) Datura

30) If the leaflets are attached at the tip of petiole, leaf is called :-

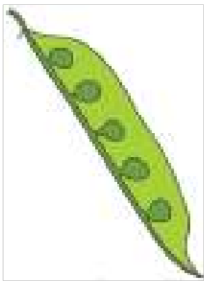
(1) Pinnately compound leaf
(2) Palmately compound leaf
(3) Simple leaf
(4) Unipinnate leaf

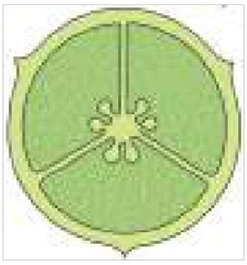
31) When the veins run parallel to each other within a lamina, the venation is termed as :-

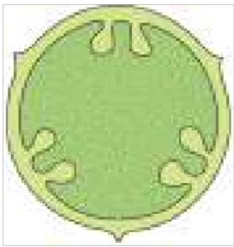
(1) Parallel
(2) Reticulate

(3) Both 1 & 2
(4) Pinnate

32) Which of the following figure represent a typical placentation as seen in mustard :-

(1) 

(2) 

(3) 

(4) 

33) A sterile stamen is called:

(1) Staminode
(2) Pistilode
(3) Epipetalous
(4) Epiphyllous

34)

In pea flowers the two lateral petals is known as:

(1) Keel
(2) Wings
(3) Standard
(4) Carina

35) Shape of corolla may be

(1) Tubular
(2) Bell-shaped
(3) Funnel shaped
(4) All of above

SECTION-2

1) Match the given columns and select the **correct** option.

	Column-I		**Column-II**
(a)	Dodo	(i)	Africa
(b)	Quagga	(ii)	Mauritius
(c)	Steller's sea cow	(iii)	Russia

	a	**b**	**c**
(1)	(i)	(ii)	(iii)
(2)	(ii)	(i)	(iii)
(3)	(iii)	(i)	(ii)
(4)	(iii)	(ii)	(i)

(1) 1
(2) 2
(3) 3
(4) 4

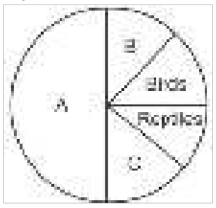
2) Which of the following is **not** a 'Evil Quartet'?

(1) Habitat loss and fragmentation
(2) Over-exploitation
(3) Co-evolution
(4) Co-extinctions

3) If S, A, Z and C are species richness, area, regression coefficient and Y-intercept respectively then, which of the given equations **correctly** represents species-area relationship as rectangular hyperbolic curve?

(1) Log A = Iog C + Z log S
(2) Log S = log A + Z log C
(3) $S = CA^Z$
(4) $C = SA^Z$

4) In the following pie chart of global vertebrate diversity, what does A, Band C represent?



	A	B	C
(1)	Mammals	Amphibians	Fishes
(2)	Amphibians	Fishes	Mammals
(3)	Fishes	Mammals	Amphibians
(4)	Amphibians	Mammals	Fishes

(1) 1
(2) 2
(3) 3
(4) 4

5) Sacred groves is also one of the important mean of biodiversity conservation find out the **odd** one with respect of sacred groves :-

(1) Sarguja, Chanda, Bastar –Uttar Pradesh
(2) Aravalli hills – Rajasthan
(3) Khasi & Jaintia hills – Meghalaya
(4) Western Ghat regions – Maharashtra and Karnataka

6) Name the family having (9) + 1 arrangement of stamens.

(1) Solanaceae
(2) Asteraceae
(3) Liliaceae
(4) Fabaceae

7) In china rose the flowers are :

(1) Epigynous with twisted aestivation
(2) Hypogynous with twisted aestivation
(3) Epigynous with valvate aestivation
(4) Hypogynous with imbricate aestivation

8) When a pair of leaves arise at node and lies opposite to each other this type of phyllotaxy found in

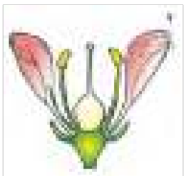
(1) Mustard
(2) *Calotropis*
(3) *Alstonia*

(4) Sunflower

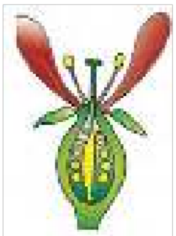
9) Mango and Coconut develops from

(1) Monocarpellary, inferior ovaries
(2) Monocarpellary, superior ovaries
(3) Multicarpellary, inferior ovaries
(4) Multicarpellary, superior ovaries

10) Which of the following type of flowers are present in Guava?

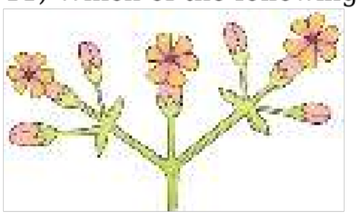
(1) 

(2) 

(3) 

(4) 

11) Which of the following is **correct** for the type of inflorescence given below ?



(1) The main axis terminates in a flower
(2) The main axis is of limited growth
(3) The flowers are borne in a basipetal order
(4) All are correct

12) Which of the floral formula is **correct** for given floral diagram ? 

(1) $\oplus$ ⚥ $K_{(5)} C_{1+2+(2)} A_{(9+1)} \underline{G}_{(2)}$

(2) % ⚥ $K_{(5)} C_{1+2+(2)} A_{(9+1)} \underline{G}_{(2)}$

(3) ⚥ $K_{2+2} C_4 A_{2+4} \underline{G}_{(2)}$

(4) $\oplus$ ⚥ $K_{2+2} C_4 A_{2+4} \underline{G}_{(2)}$

13) Which of the following is **incorrect** of stem?

(1) Develops from plumule of embryo of a germinating seed

(2) Bears nodes and internodes

(3) Bears only terminal bud not axillary bud

(4) Main function of stem is spreading out branches bearing leaves, flowers and fruits.

14) **Statement-I :-** In majority of the dicotyledonous plants, the direct elongation of the radicle leads to the formation of primary roots.
Statement-II :- In monocotyledonous plants, the primary root is long lived and is not replaced by large number of roots.

(1) Both **Statement I** and **Statement II** are incorrect.

(2) **Statement I** is correct but **Statement II** is incorrect.

(3) **Statement I** is incorrect but **Statement II** is correct

(4) Both **Statement I** and **Statement II** are correct.

15) Which of the following statement is **correct** about dicot seed:
(a) Micropyle is present above the hilum
(b) At the two ends of the embryonal axis are present the radicle and the plumule
(c) Seed coat has three layers
(d) Cotyledons are often fleshy and without reserve food materials.

(1) a and d only

(2) a and b only

(3) b and c only

(4) c and d only

ZOOLOGY

SECTION-1

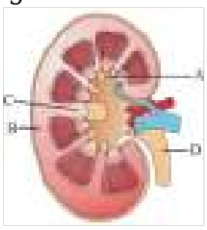
1) Which of the following statements is **false**-
I. Outer cortex and inner medulla are the two zones in kidney
II. Medulla is divided into renal pyramids
III. Pyramid projects into calyx
IV. Inwards extension of cortex between the pyramids is called renal column of Bertini

(1) I and IV
(2) II and IV
(3) IV
(4) None

2) In micturition, signal is initiated by :-

(1) CNS
(2) Stretching of urinary bladder
(3) Stretching of internal urethral sphincter
(4) Stretch receptors on urethra

3) Given is the figure of longitudinal section of kidney. Identify the parts labelled as A to D and select the correct option :-



	A	B	C	D
(1)	Cortex	Calyx	Renal column	Ureter
(2)	Calyx	Cortex	Renal column	Ureter
(3)	Medulla	Cortex	Renal capsule	Urethra
(4)	Calyx	Cortex	Renal pelvis	Urethra

(1) 1
(2) 2
(3) 3
(4) 4

4) The maximum amount of electrolytes and water (70-80 percent) from the glomerular filtrate is reabsorbed in which part of the nephron ?

(1) Ascending limb of loop of Henle
(2) Distal convoluted tubule
(3) Proximal convoluted tubule
(4) Descending limb of loop of Henle

5) Which of the following is the correct pathway for passage of urine in humans ?

(1) Collecting tubule → Ureter → Bladder → Urethra
(2) Renal vein → Renal ureter → Bladder → Urethra
(3) Pelvis → cortex → Bladder → Urethra
(4) Cortex → Medulla → Bladder → Ureter

6) A large quantity of fluid is filtered everyday by nephrons in the kidneys. Only about 1% of it is excreted as urine. The remaining 99% of the filtrate is

(1) Stored in the urinary bladder
(2) Reabsorbed into the blood
(3) Gets collected in the renal pelvis
(4) Lost as sweat

7) Which of the following sequences is correct regarding regulation of kidney function?

(1) An excess loss of water from body → Hypothalamus → Osmoreceptors → Neurohypophysis → ADH → Increases water permeability of DCT and CT → Prevention of diuresis
(2) An excess loss of fluid from body → Osmoreceptors → Hypothalamus → Neurohypophysis → ADH → Increases water permeability of DCT and CT → Prevention of diuresis
(3) An excess loss of fluid from body → Osmoreceptors → Hypothalamus → Neurohypophysis → Aldosterone → Water permeability of DCT and CT increases → Prevention of diuresis
(4) An excess loss of fluid from body → Osmoreceptors → Hypothalamus → Adenohypophysis → ADH → Increases water permeability of DCT and CT → Prevention of diuresis

8) Select the correct option representing the parts of nephron that respectively reabsorb (i) glucose, (ii) amino acids, (iii) inorganic ions (Na^+, K^+, Cl^-) and (iv) urea.

	(i)	(ii)	(iii)	(iv)
(1)	DCT	Descending limb of loop of Henle	DCT	DCT
(2)	DCT	Descending limb of loop of Henle	PCT	DCT
(3)	PCT	PCT	PCT	Collecting duct
(4)	PCT	DCT	DCT	Ascending limb of loop of Henle

(1) (1)
(2) (2)
(3) (3)
(4) (4)

9) Dialyzing unit (artificial kidney) contains a fluid which is almost same as plasma except that it has

(1) High glucose
(2) High urea
(3) No urea

(4) High uric acid

10) **Statement I:** Movement is one of the significant features of living beings.
Statement II: Streaming of protoplasm in the unicellular organisms like Amoeba is a simple form of movement.

(1) Both statements I and II are correct
(2) Both statements I and II are incorrect
(3) Only statement I is correct
(4) Only statement II is correct

11) pH of urine is -

(1) Highly acidic
(2) Slightly acidic
(3) Highly alkaline
(4) Slightly alkaline

12) Uric acid is the main excretory product in :

(1) Insects
(2) Earthworm
(3) Amphibians
(4) Mammals

13) Renal calculi refers to

(1) Insoluble mass of crystallised salts formed within the kidney
(2) Soluble mass of non-crystallised salts formed within the kidney
(3) Insoluble mass of non-crystallised salts formed within the kidney
(4) Soluble mass of crystallised salts formed within the kidney

14) An adult human excreates on an average

(1) 2–3 litres of urine per day
(2) 1–1.5 litres of urine per day
(3) 2–5 litres of urine per day
(4) 4–5 litres of urine per day

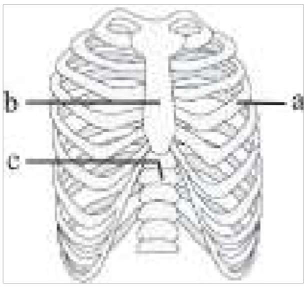
15) Which of the following is responsible for excretion of dilute urine ?

(1) More secretion of insulin
(2) Less secretion of vasopressin
(3) More secretion of aldosterone
(4) Less secretion of glucagon

16)

A Malpighian body is constituted by

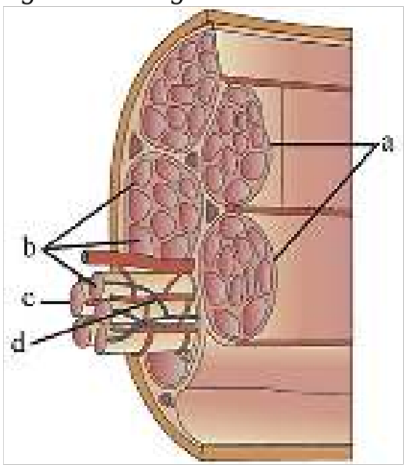
(1) Glomerulus only
(2) Glomerulus and Bowman's capsule
(3) Glomerulus and efferent vessel
(4) Glomerulus, Bowman's capsule and efferent vessel

17) Identify a, b, c . 

(1) a-sternum, b-ribs, c-vertebral column
(2) a-sternum, b-vertebral column, c-ribs
(3) a-vertebral column, b-sternum, c-ribs
(4) a-ribs, b-sternum, c-vetebral column

18)

Recognise the figure and find out the correct matching :-



(1) a–muscle fibre, b-muscle cell, c-fascicle, d–muscle bundle
(2) a-fascicle, b–muscle fibre, c-sarcolemma, d–blood capillary
(3) a–muscle bundle, b-muscle cell, c-sarcolemma, d–blood capillary
(4) Both (2) and (3)

19) Match column I and column II and choose correct option from followings :-

	Column-I		Column-II

A	H-Zone	i	Actinin protein
B	Sarcoplasmic Reticulum	ii	Light central area of A-band.
C	Z-line	iii	Part of meromyosin protein.
D	Cross Arm	iv	Store house of Ca^{+2} ions

(1) A-ii, B-iii, C-i, D-iv
(2) A-iii, B-ii, C-iv, D-i
(3) A-ii, B-iv, C-i, D-iii
(4) A-i, B-iii, C-iv, D-ii

20) Muscle bundles or fascicles held together by a common collagenous tissue layer called :-

(1) Motor end plate
(2) Motor unit
(3) Fascia
(4) Neuromuscular junction

21) Match the Column-I and Column-II and select the correct option ?

	Column I		**Column II**
(A)	Tetany	(i)	Paralysis of skeletal muscle
(B)	Muscular dystrophy	(ii)	Inflammation of joints
(C)	Arthritis	(iii)	Degeneration of skeletal muscles
(D)	Myasthenia gravis	(iv)	Rapid spasm in muscle

(1) B-i, A-ii, C-iii, D-iv
(2) A-iv, B-iii, C-ii, D-i
(3) D-iii, B-ii, C-i, A-iv
(4) A-iv, C-ii, B-i, D-iii

22) Which pairs of ribs are known as False ribs ?

(1) 7^{th}, 8^{th} and 9^{th} pairs
(2) 8^{th}, 9^{th} and 10^{th} pairs
(3) First 3 pairs of ribs
(4) 6^{th}, 7^{th} and 8^{th} pairs

23) The contractile protein of skeletal muscle involving ATPase activity is :-

(1) Actin
(2) Myosin
(3) Troponin
(4) Tropomyosin

24) Match the appropriate number of the bones associated with given structures and choose the correct option

a.	Limb bones	(i)	12
b.	Thoracic vertebrae	(ii)	2
c.	Hip girdle	(iii)	30
d.	Lumbar vertebrae	(iv)	5
		(v)	3

(1) a-(iii), b-(i), c-(ii), d-(iv)
(2) a-(iii), b-(i), c-(v), d-(iv)
(3) a-(iv), b-(i), c-(ii), d-(iii)
(4) a-(iii), b-(ii), c-(i), d-(iv)

25) **Assertion :** All locomotions are movement but all movements are not locomotions.
Reason : Some voloutary movements results in a change of place or location are only called locomotion.

(1) Both Assertion and Reason are true and Reason is the correction explanation of Assertion.
(2) Both Assertion and Reason are true and Reason is not the correction explanation of Assertion.
(3) Assertion is true but Reason is false.
(4) Both Assertion and Reason are false.

26) **Assertion :** Muscles have special properties like excitability, contractility, extensibility and elasticity.
Reason : Muscles are used for changes in body postures and locomotion as well

(1) Both **Assertion** and **Reason** are true but **Reason** is NOT the correct explanation of **Assertion**.
(2) **Assertion** is true but **Reason** is false.
(3) **Assertion** is false but **Reason** is true.
(4) Both **Assertion** and **Reason** are true and **Reason** is the correct explanation of **Assertion**.

27) Joint between head of humerus and glenoid cavity of scapula is :-

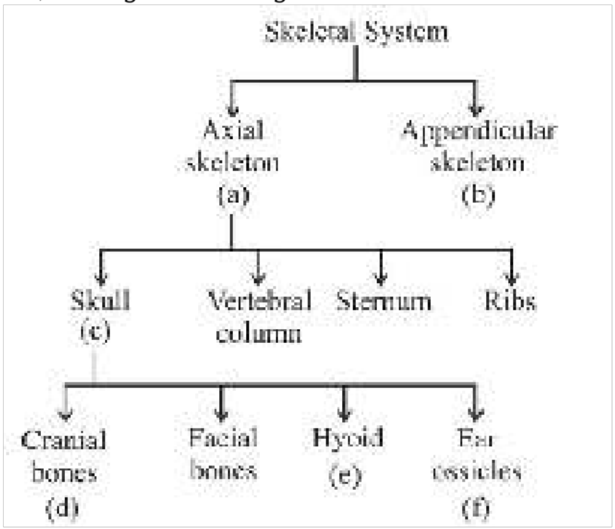
(1) Ball and socket
(2) Hinge
(3) Pivot
(4) Condylar

28) Find out the correct order of number of bones in the parts of forelimb such as humerus, radius,

ulna, Carpals, metacarpals and phalanges :-

(1) 1, 1, 1, 4, 9, 14
(2) 1, 1, 1, 6, 11, 14
(3) 1, 1, 1, 8, 5, 14
(4) 1, 1, 8, 1, 5, 14

29) Recognise the figure and find out the correct matching w.r.t number of bones :-



(1) a—126, b—80, c—14, d—6, e—8, f—3
(2) a—80, b—126, c—29, d—14, e—1, f— 3
(3) a—80, b—126, c—29, d—8, e—1, f—6
(4) a—126, b—80, c—29, d—14, e—2, f—6

30) **Statement I:** The peripheral nervous system (PNS) includes all the nerves of the body that are not part of the central nervous system (CNS).
Statement II: The visceral nervous system is a separate nervous system that operates independently of the peripheral nervous system (PNS).

(1) Statement I is false but Statement II is true
(2) Both Statement I and Statement II are true
(3) Both Statement I and Statement II are false
(4) Statement I is true but Statement II is false

31) **Statement I:** Dendrites are short fibres that contain Nissl's granules and transmit impulses towards the cell body.
Statement II: The axon is a short fibre that transmits impulses away from the cell body and terminates in a structure called the synaptic knob.

(1) Statement I is false but Statement II is true
(2) Both Statement I and Statement II are true
(3) Both Statement I and Statement II are false
(4) Statement I is true but Statement II is false

32) Given below is the list of brain structures. Find out the correct location and functions of these

structures :-

Structure		Location	Functions
(1)	Hypothalamus	Hind brain	Motor signaling
(2)	Corpus callosum	Inner cerebral hemisphere	Connects both cerebral hemisphere
(3)	PONS	Optic lobes	Respiration and glastric control
(4)	White matter	Cerebrum dorsal surface	myelinated Axon

(1) 1
(2) 2
(3) 3
(4) 4

33)

Column-I			Column-II
1.	Unipolar neuron	i.	Found in the cerebral cortex
2.	Myelinated Nerve fibres	ii.	Found usually in the embryonic stage
3.	Multipolar neuron	iii.	Commonly found in autonomous and the somatic neural systems
4.	Unmyelinated nerve fibre	iv.	Found in spinal and cranial nerves.

(1) 1-ii, 2-iv, 3-i, 4-iii
(2) 1-ii, 2-iii, 3-i, 4-iv
(3) 1-i, 2-iv, 3-ii, 4-iii
(4) 1-i, 2-iii, 3-ii, 4-iv

34) Medulla controls :-

(1) Respiration
(2) Gastric secretions
(3) Cardiovascular reflexes
(4) All of the above

35)

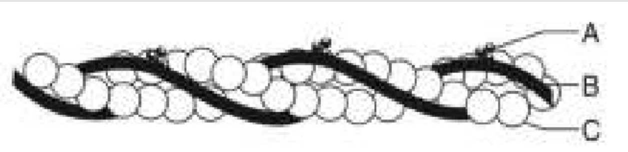
Na^+ K^+ pump transports?

(1) 2 Na^+ influx, $3K^+$ efflux
(2) $3Na^+$ influx, $2K^+$ efflux
(3) $3Na^+$ efflux, $2K^+$ influx
(4) $3Na^+$ efflux, K^+ influx

SECTION-2

1)

Following is the figure of actin (thin) filaments. Identify A, B and C.



(1) A-Tropomyosin, B-Troponin, C- F actin

(2) A-Tropomyosin, B-Myosin, C- Tropomyosin

(3) A-Troponin, B-Tropomyosin, C- G-actin

(4) A-Troponin, B-Tropomyosin, C- F actin

2) Given below are two statements:
Statement I: A synapse is formed by the membranes of a pre-synaptic neuron and a post-synaptic neuron, which are always separated by a gap called synaptic cleft.
Statement II: There are two types of synapses, namely, electrical synapses and chemical synapses. In the light of the above statements, choose the correct answer from the options given below:

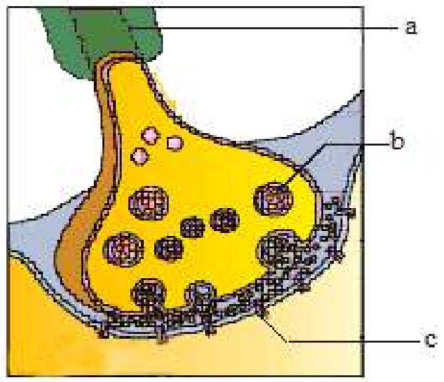
(1) Statement I is true but Statement II is false

(2) Both Statement I and Statement II are true.

(3) Both Statement I and Statement II are false

(4) Statement I is false but Statement II is true

3) Label the parts shown in the below figure correctly and choose the correct option accordingly :-



(1)	a – Dendron	b – Synapse	c – vesicle
(2)	a – Axon	b – Synaptic vesicles	c – Receptors
(3)	a – Axon	b – Synaptic cleft	c – Neurotrans mitter
(4)	a – Synapse	b – Receptor	c – Dendron

(1) 1

(2) 2

(3) 3

(4) 4

4) Fibres tract of different region of brain interconnect at:-

(1) Medulla
(2) Hypothalamus
(3) Thalamus
(4) Pons

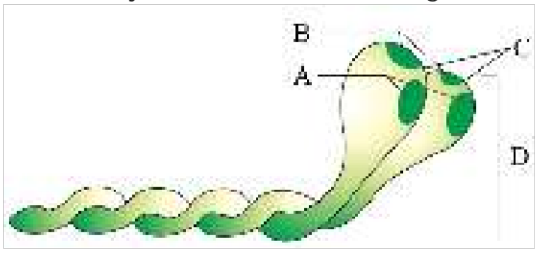
5) The inner parts of cerebral hemispheres and a group of associated deep structures like amygdala, hippocampus, etc; form a complex structure called :-

(1) Recticular system
(2) Corpora quadrigemina
(3) Limbic lobe/limbic system
(4) Arbor vitae

6) Unmyelinated nerve fibre is enclosed by a Schwann cell that does not form a myelin sheath around the axon and is commonly found in ?

(1) Autonomus neural system
(2) Central neural system
(3) Somatic neural system
(4) Both 1 & 3

7) Identify A, B, C and D in the given diagram and choose the **correct** option.



(1) A-ATP binding site; B-Actin binding site; C-Head; D-Tail
(2) A-ATP binding site; B-Actin binding site; C-Head; D-Cross arm
(3) A-ATP binding site; B-Myosin binding site; C-Head; D-Cross arm
(4) A-ATP enzyme; B-Actin binding site; C-Cross arm; D-Head

8) Contraction of muscle fibres takes place by the sliding of:

(1) Thin filaments over the actin filaments
(2) Thick filaments over the thin filaments
(3) Thin filaments over the thick filaments
(4) Thick filaments over the myosin filaments

9) Which one of the following represents the correct order of vertebral regions from superior to inferior ?

(A) Sacrum (B) Thoracic
(C) Cervical (D) Lumbar
(E) Coccyx

(1) C → B → D → A → E

(2) A → B → C → D → E

(3) C → D → B →A → E

(4) B → A → D → C → E

10) Acromian process is part of :-

(1) Vertebral column

(2) Pelvic girdle

(3) Femur

(4) Pectoral girdle

11) Glomerular filtration occurs through filtration membrane which contains all **except**

(1) Epithelial cells of Bowman's capsule

(2) Epithelial cells of glomerular blood capillaries

(3) Epithelial cells of efferent arteriole

(4) Basement membrane

12) **Assertion :** White fibres are also called aerobic muscles.
Reason : They contain plenty of mitochondria and utilise oxygen for ATP production.

(1) Both assertion and reason are true and reason is the correct explanation of assertion.

(2) Both assertion and reason are true but reason is not the correct explanation of assertion.

(3) Both assertion and reason are false.

(4) Assertion is true but reason is false.

13) Which of the following is not correct with respect to synovial joints ?

(1) Presence of fluid filled cavity between the articulating surfaces.

(2) The joint between adjacent vertebrae in the vertebral column is of this pattern.

(3) They allows considerable movement.

(4) Saddle joint is an example of synovial joint.

14) Amoeba shows movement with the help of

(1) Pseudopodia

(2) Flagella

(3) Cilia

(4) Muscle

15) What is the name of the process in which muscles produce energy anaerobically, leading to fatigue?

(1) Glycolysis

(2) Krebs cycle

(3) Lactic acid formation

(4) Oxidative phosphorylation

ANSWER KEYS

PHYSICS

SECTION-1

Q.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
A.	4	4	3	4	4	2	4	1	3	1	3	1	2	1	2	3	4	1	3	2
Q.	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35					
A.	4	2	4	2	1	3	4	1	1	2	2	1	2	3	2					

SECTION-2

Q.	36	37	38	39	40	41	42	43	44	45	46	47	48	49	50
A.	1	1	1	3	1	1	2	4	1	1	2	2	3	3	1

CHEMISTRY

SECTION-1

Q.	51	52	53	54	55	56	57	58	59	60	61	62	63	64	65	66	67	68	69	70
A.	4	2	1	3	3	2	3	1	2	4	2	4	2	1	1	1	2	2	2	4
Q.	71	72	73	74	75	76	77	78	79	80	81	82	83	84	85					
A.	2	2	4	4	4	4	1	2	4	2	3	4	1	3	4					

SECTION-2

Q.	86	87	88	89	90	91	92	93	94	95	96	97	98	99	100
A.	1	1	1	3	1	1	3	2	3	4	2	2	1	2	1

BOTANY

SECTION-1

Q.	101	102	103	104	105	106	107	108	109	110	111	112	113	114	115	116	117	118	119	120
A.	2	2	2	3	3	1	4	1	3	1	2	4	3	2	1	2	3	1	2	1
Q.	121	122	123	124	125	126	127	128	129	130	131	132	133	134	135					
A.	4	1	1	2	1	2	4	3	3	2	1	3	1	2	4					

SECTION-2

Q.	136	137	138	139	140	141	142	143	144	145	146	147	148	149	150
A.	2	3	3	3	1	4	2	2	2	3	4	4	3	2	2

ZOOLOGY

SECTION-1

Q.	151	152	153	154	155	156	157	158	159	160	161	162	163	164	165	166	167	168	169	170
A.	**4**	**2**	**2**	**3**	**1**	**2**	**2**	**(3)**	**3**	**1**	**2**	**1**	**1**	**2**	**2**	**2**	**4**	**4**	**3**	**3**

Q.	171	172	173	174	175	176	177	178	179	180	181	182	183	184	185
A.	**2**	**2**	**2**	**1**	**1**	**1**	**1**	**3**	**3**	**4**	**4**	**2**	**1**	**4**	**3**

SECTION-2

Q.	186	187	188	189	190	191	192	193	194	195	196	197	198	199	200
A.	**4**	**4**	**2**	**4**	**3**	**4**	**2**	**3**	**1**	**4**	**3**	**3**	**2**	**1**	**3**

SOLUTIONS

PHYSICS

4)

stress = $\frac{F}{Area} \propto \frac{1}{r^2}$

$\frac{\rho_1}{\rho_2} = \left(\frac{r_2}{r_1}\right)^2$

9)

$\omega = \omega_0 + \alpha t$

$\omega = 1.5 \times 20 = 30$ rad/s

$v = r\omega = 0.4 \times 30$

$v = 12$ m/s

11) conceptual

15) $\Delta L = \frac{F(L/2)}{AY} = \frac{(AL\rho g)(L/2)}{AY} = \frac{\rho g L^2}{2Y}$

18) Slope of the curve stress-strain represents young's Modulus. and Y depends upon temp. if temp increases, Y decreases so $Y_2 > Y_1$ then $T_1 > T_2$

25) A body which regains its original shape after the removal of external force is called elastic body. i – q

A body which does not regain its original shape after the removal of external force is called plastic body.

ii - r

A body which does not show any deformation on applying external forces is called rigid body.

iii - s

The property of the body to regain its original configuration when the deforming force are removed is called elasticity.

iv - p

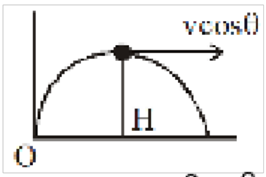
26) Gravitational field intensity inside the hollow sphere will be zero.

So, no work will be done inside the sphere to move it.

So, $V_{in} = V_{surface}$ (potential will remain same)

$= -20$ J/kg

27) L = mvcosθ H



$L = mv\cos\theta \dfrac{v^2\sin^2\theta}{2g}$

$L = \dfrac{mv^3\sin^2\theta\ \cos\theta}{2g}$

29) Inside a spherical shell, gravitational field is zero and hence potential remains same everywhere

31)

$I(3\omega) + 2I\omega = (I + 27)\,\omega'$

$\dfrac{5I\omega}{3I} = w'$

$w' = \dfrac{5}{3}\omega$

32) $F = \dfrac{Gm_1m_2}{r^2}$

$\Rightarrow F \propto \dfrac{1}{r^2}$

$\Rightarrow F \propto m_1m_2$

⇒ This force provides centripetal force and acts towards sun

$\Rightarrow T^2 \propto a^3$ (Kepler's third law)

34) $E = \dfrac{GMm}{2r}$

$L = mv^{0}r = m\sqrt{\dfrac{GM}{r}}r$

$L = \sqrt{\dfrac{GMm^2r^2}{r}}$

$L = \sqrt{\dfrac{2GMm}{2r} \times mr^2}$

$L = \sqrt{2E\,mr^2}$

35) $Y = \dfrac{\frac{F}{A}}{\frac{\Delta\ell}{\ell}} \Rightarrow \Delta\ell = \dfrac{F\ell}{AY}$

But V = A□ so $A = \dfrac{V}{\ell}$ (V = volume)

Therefore $\Delta\ell = \dfrac{F\ell^2}{VY} \propto$ □^{2}

38) $P = \tau \cdot \omega$

$P = I\ \alpha \cdot \omega = I\left(\omega \frac{d\omega}{d\theta}\right) \cdot \omega$

$\omega^2 d\omega = \frac{P}{I} d\theta$

$\omega \propto \theta^{1/3}$

$\omega \propto (n)^{1/3}$ $[\theta = 2\pi n]$

41) $a = \frac{3m - m}{3m + m} g = \frac{g}{2}$

Acceleration of centre of mass $= \frac{-3m \times \frac{g}{2} + \frac{mg}{2}}{3m + m} = \frac{-g}{4}$ (here – represent downward direction)

42) $T = \frac{2\pi R}{v}$

$\frac{GMm}{R^n} = \frac{mv^2}{R}$

or $v = \left[\frac{GM}{R^{n-1}}\right]^{1/2}$

$T = \frac{2\pi R}{\sqrt{GM/R^{n-1}}} = \frac{2\pi}{\sqrt{GM}} \times R^{(n+1)/2}$

$\therefore \quad T \propto R^{(n+1)/2}$

44)

$\theta = 2t^2 + 3t + 4$

$\omega = 4t + 3$

$\alpha = 4$

(P) $\omega_{t=2} = 4 \times 2 + 3 = 11$

(Q) $\alpha_{t=3} = 4$

(R) $\langle\omega\rangle = \frac{\theta_2 - \theta_0}{2}$

$= \frac{(2 \times 4 + 3 \times 2 + 4) - (4)}{2} = 7$

(S) $\langle\alpha\rangle = \frac{\omega_2 - \omega_0}{2} = \frac{11 - 3}{2} = 4$

45)

Ex-1 Q.No.-82

50)

$I = m R^2/2$

$\frac{19I}{6} =$

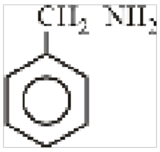
58) Br_2 is reduced to HBr.

59) $M^{3+} \rightarrow M^{6+} + 3e^-$.

60) $CaOCl_2$ is actually $Ca^{2+}(O\overset{+1}{Cl}^-)\overset{-1}{Cl^-}$. Therefore, the O.N. of the two Cl atoms are + 1 and –1.

61) In reaction (b), P_4 is both reduced to PH_3 and oxidised to NaH_2PO_2. Thus, it is a disproportionation reaction. But in all other redox reactions, different species are oxidised/reduced, i.e., in reaction (a), Mg is oxidised but N_2 is reduced; in reaction, (c) Cl_2 is reduced but KI is oxidised to I_2 in reaction (d). Cr_2O_3 is reduced but Al is oxidised.

66)

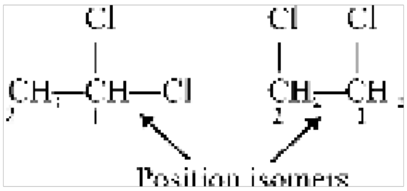
Duo to localized lone pair  is the strongest base

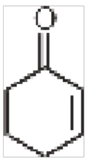
69)

Stability of $C^{\oplus} \propto$ $pK_a \propto \frac{1}{K_a}$
Stability of $C^{o} \propto$ +M / Resonance / +H.C. / +I
Stability of $C^{-} \propto$ –M / Resonance / –I

70)

Fact

71) 

81) 

84) NCERT (XI) Pg. # 357 (Prob. No. 12.21)

86) W = 0.5

Normality of H_2SO_4 → N = Molarity × Valence factor

$$= \frac{1}{2} \times 2 = 1$$

Volume of H_2SO_4, used in neutralising NH_3 → V → 20 ml

% of Nitrogen → $\dfrac{1.4 \times N \times V}{w}$

→ $\dfrac{1.4 \times 1 \times 20}{0.5}$

→ 56 %

87)

Na_2S + conc. $HNO_3 \rightarrow H_2S\uparrow$

NaCN + conc. $HNO_3 \rightarrow$ HCN

93) Answer (2)

–CN and – NC act as different functional groups, therefore RCN and RNC are functional isomers.

95)

Z-isomers have senior groups on same side.

BOTANY

132) NCERT Pg. # 65

134)

NCERT-XI, Pg # 70

Chinarose is a dicot plant and most leaves of dicotyledonous plants posses reticulate venation.

145) NCERT-XI, Pg. # 62

146)

NCERT-XI, Pg. # 62

147)

NCERT-XI, Pg # 67

150) NCERT Pg # 76

152) NCERT-XI, Pg. # 297

154)

NCERT Page # 209

156) NCERT Pg.no 293.

157) NCERT Pg.no 297.

158) Ncert.

159) NCERT Pg.no 298.

163) NCERT Pg. # 299

164) NCERT (XIth) Pg. # 296, 297

165) NCERT (XIth) Pg. # 295

167) NCERT XI Pg. # 310 Fig. 20.8

168)

NCERT Based.

171) NCERT (XIth) Pg. # 312

172) NCERT (XIth) Pg. # 310

173)

NCERT - Pg. No. 221 Para 2 Fig 17.3 (b)

174)

NCERT Pg. No. 241

176) NCERT-XI, Page No. 304

180) Statement I: Correct. The peripheral nervous system (PNS) comprises all the nerves of the body that are associated with the central nervous system (CNS), which includes the brain and spinal cord. Statement II: Incorrect. The visceral nervous system is not a separate system but rather a part of the peripheral nervous system that deals with the innervation of the viscera, conveying impulses to and from the central nervous system.

181) Statement I: Correct. Dendrites are indeed short fibres that contain Nissl's granules and are responsible for transmitting impulses towards the cell body. Statement II: Incorrect. The axon is not a short fibre, but rather a long one that transmits impulses away from the cell body and terminates in a structure called the synaptic knob, which contains synaptic vesicles with neurotransmitters.

183) NCERT Pg. # 317(E), 318 (H)

186) NCERT Pg. # 317, para 1

187) Statement I: Incorrect. A synapse may or may not be separated by a gap known as the synaptic cleft. In the case of electrical synapses, the pre-synaptic and post-synaptic neurons are connected via gap junctions that allow direct passage of ions and signaling molecules, meaning there is no gap. Statement II: Correct. There are indeed two types of synapses: electrical synapses and chemical synapses, which use different mechanisms for transmitting signals between neurons.

188)

a - Axon, b - Synaptic vesicles, c - Receptors

192)

Old, NCERT Page no. 306

193)

Old, NCERT Page no. 307

197) NCERT XI Page # 223

198) NCERT XI Page # 227

199) NCERT Page No. 217

200) NCERT-XI, Page # 223